Non-Inverting | Inverting Amplifiers

How do opamps work?

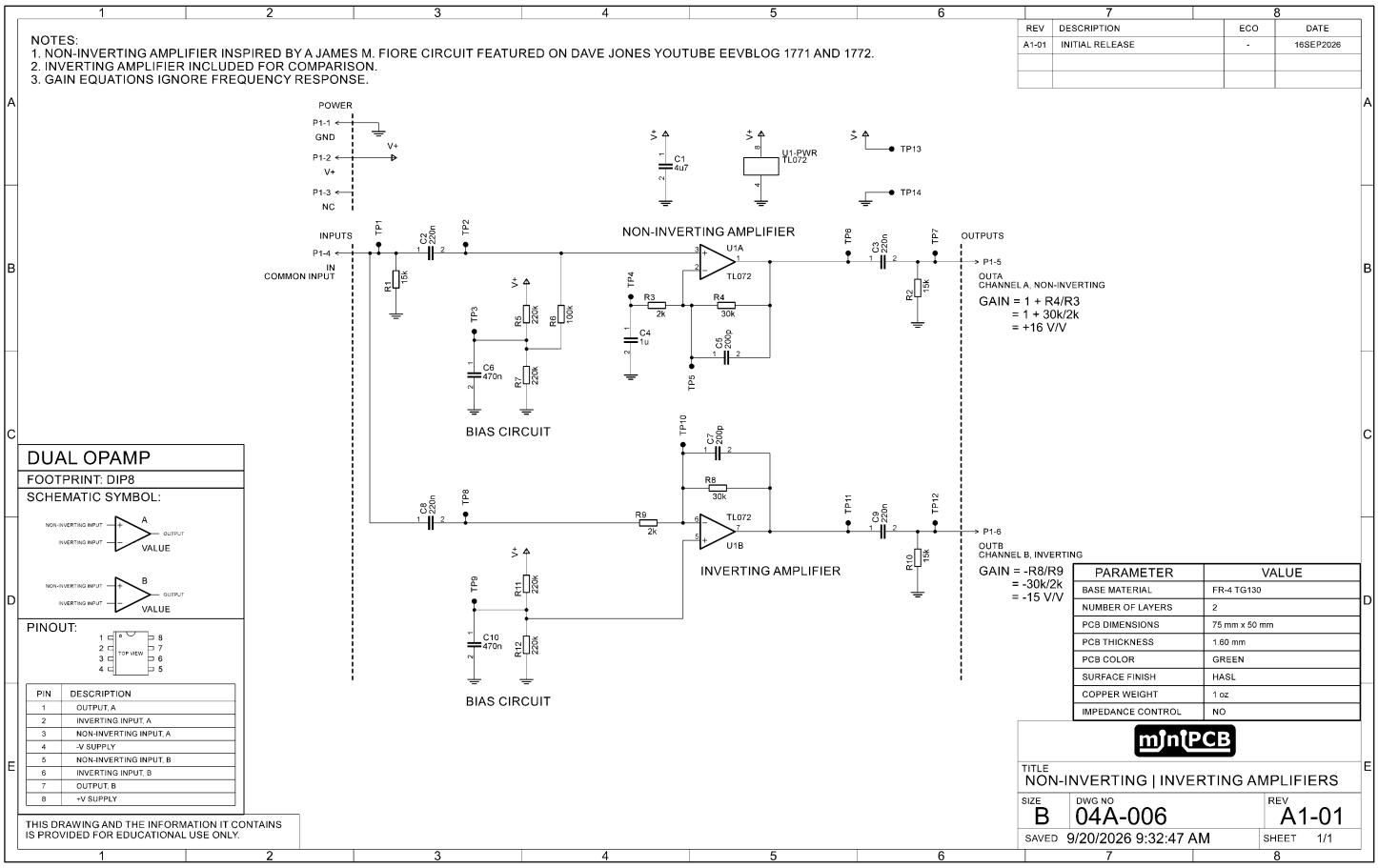
An opamp will adjust its output as needed so that, when negative feedback is present and the amplifier is operating within its limits, the voltage at the inverting input is approximately equal to the voltage at the non-inverting input.

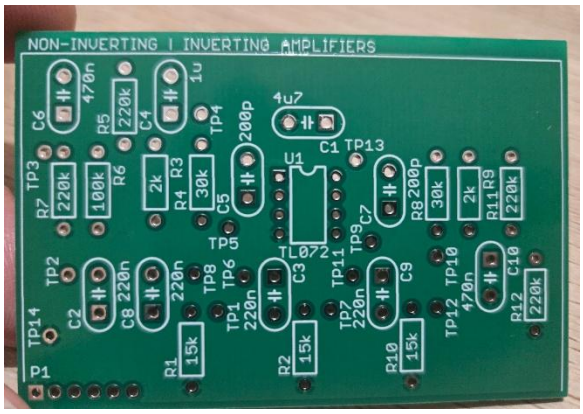
How do resistors work?

A resistor opposes the flow of current, creating a voltage drop that is proportional to the current flowing through it. The relationship is described by Ohm's law: V = I × R

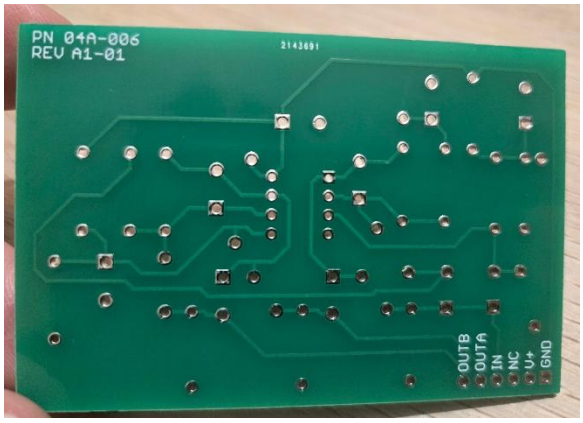
How do capacitors work?

A capacitor stores electrical energy in an electric field and resists changes in voltage by charging or discharging as needed. The faster the voltage changes, the more current flows through the capacitor.





Front Side



Back Side

miniPCB Part Number

PART NO	TITLE	PIECES PER PANEL
04A-006	Non-Inverting | Inverting Amplifiers	1

miniPCB Revision History

REV	DESCRIPTION	DATE
A1-01	Initial Release	16SEP2026

LLM Interaction

If you paste this link into a chat with an LLM that can access public GitHub repositories, you should be able to chat about this miniPCB with your favorite AI!

https://github.com/miniPCB/minipcb.github.io/blob/main/md/04A-006_A1-01_sch.md

See how smart your LLM is! Can it recommend part value ranges for each component? Suggest manufacturers and manufacturer part numbers? Inform you of all the part manufacturers and cite and provide links to documentation? Tell you the current BOM cost for given component distributors and a pricing breakdown showing various manufacturing quantities? Generate an assembly procedure? Generate a test procedure? Write a lab report based on a spreadsheet of data and a folder with scope shots? How good is its visibility of the industry? For example, does it know the industry's history, current investors/leaders, religions (like signal integrity or power integrity) and more?

Pinout

REF DES	SIGNAL DESCRIPTION
P1-1	Power supply common return, 0V
P1-2	Power supply positive source, +5V
P1-3	No connection
P1-4	AC signal input, 15k ohm load resistor
P1-5	AC signal non-inverting output, 15k ohm load resistor
P1-6	AC signal inverting output, 15k ohm load resistor

Test Points

REF DES	SIGNAL DESCRIPTION
TP1	Measure input signal, DC coupled
TP2	Measure input signal, AC coupled, channel A
TP3	Measure DC bias, channel A
TP4	Measure feedback signal at AC coupling feedback capacitor, channel A
TP5	Measure feedback signal, channel A
TP6	Measure output signal, DC coupled, channel A
TP7	Measure output signal, AC coupled, channel A
TP8	Measure input signal, AC coupled, channel B
TP9	Measure DC bias, channel B
TP10	Measure feedback signal, channel B
TP11	Measure output signal, DC coupled, channel B
TP12	Measure output signal, AC coupled, channel B
TP13	Measure supply input, V+
TP14	Common signal return for all measurements

Circuit Description

This miniPCB implements two operational-amplifier circuits for direct comparison: a non-inverting amplifier on Channel A and an inverting amplifier on Channel B. Both circuits are implemented using a TL072 dual operational amplifier (U1), operate from a single +5 V supply, and process the same externally applied input signal. The board provides numerous test points so that the input, bias, feedback, and output signals can be observed at different locations within each circuit.

The nominal AC voltage gain of Channel A is approximately +16, while the nominal AC voltage gain of Channel B is approximately −15. This allows the two amplifier configurations to be compared using nearly the same signal amplitude while clearly demonstrating the polarity difference between non-inverting and inverting operation. The gain equations describe the nominal closed-loop gain and do not account for frequency-dependent effects.

Power Supply Conditioning

The board operates from a single +5 V supply connected through P1. C1 is a 4.7 µF power supply filter capacitor located near the positive supply connection of U1 and provides local supply decoupling for the operational amplifier.

TP13 provides access to the positive supply rail, while TP14 provides a common ground reference for measurements throughout the board.

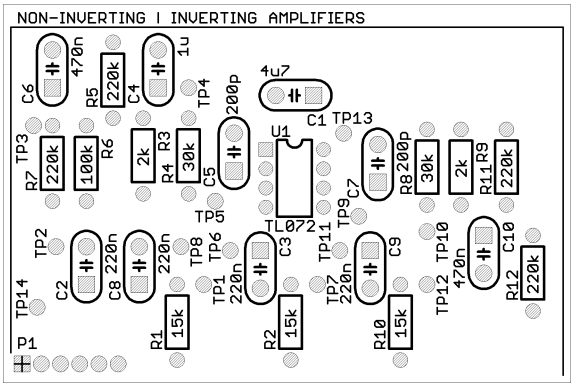


Figure 1 - Single Board, Component Outlines

Input Signal

The input signal enters the board through P1 and is shared by both amplifier channels. R1 provides a 15 kΩ load from the input to ground. TP1 allows the applied input signal to be measured before the signal is AC coupled into either amplifier.

C2 AC couples the input signal into Channel A, while C8 performs the same function for Channel B. These capacitors block the DC component of the externally applied signal and allow each amplifier circuit to establish its own internal DC operating point.

TP2 provides access to the AC-coupled Channel A input, and TP8 provides access to the AC-coupled Channel B input.

DC Bias Networks

Because the TL072 is operated from a single supply, both amplifier channels use an internal DC bias reference approximately halfway between V+ and ground.

For Channel A, R5 and R7 form a 220 kΩ / 220 kΩ voltage divider. C6 filters the resulting bias reference, and R6 connects the AC-coupled input signal to this DC reference. TP3 provides direct access to the Channel A bias voltage.

Channel B uses a similar bias network formed by R11 and R12, with C10 providing filtering. This reference is connected directly to the non-inverting input of the Channel B operational amplifier. TP9 provides access to the Channel B bias voltage.

With a nominal +5 V supply and equal-value divider resistors, both bias networks establish a nominal DC reference of approximately +2.5 V.

Channel A, Non-Inverting Amplifier

Channel A is configured as a non-inverting amplifier. The AC-coupled input signal is applied to the non-inverting input of U1A through C2 and is biased to the Channel A DC reference through R6.

The feedback network consists primarily of R4 and R3. R4 is the high-side feedback resistor connected between the amplifier output and the inverting input, while R3 is the low-side gain-setting resistor. C4 AC couples R3 to ground.

For AC signals above the effective cutoff frequency of the feedback network, the nominal closed-loop gain is:

Channel A Gain = 1 + R4 / R3

Using the installed values:

Channel A Gain = 1 + 30 kΩ / 2 kΩ = +16

Because C4 blocks DC, the amplifier does not apply the same gain of 16 to the DC bias voltage. This allows the amplifier output to remain centered near the internal bias voltage while providing the intended higher gain for AC signals.

C5, a 200 pF capacitor connected across the high-side feedback path, provides high-frequency feedback compensation. TP4 and TP5 provide access to different locations within the feedback network, and TP6 provides access directly to the DC-coupled output of U1A.

Channel B, Inverting Amplifier

Channel B is configured as an inverting amplifier. The AC-coupled input signal passes through C8 and then through R9 to the inverting input of U1B. The non-inverting input is held at the Channel B DC bias reference.

R8 provides feedback from the amplifier output to the inverting input. Together, R8 and R9 establish the nominal closed-loop AC gain:

Channel B Gain = −R8 / R9

Using the installed values:

Channel B Gain = −30 kΩ / 2 kΩ = −15

The negative sign indicates that the Channel B output is inverted relative to the input signal. C7, a 200 pF capacitor connected across the high-side feedback path, provides high-frequency feedback compensation.

TP10 provides access to the Channel B feedback node, while TP11 provides access directly to the DC-coupled output of U1B.

Output Coupling

The direct outputs of both operational amplifiers are centered around their respective DC bias levels. The board therefore provides both DC-coupled measurement points and AC-coupled external outputs.

For Channel A, TP6 allows the output of U1A to be measured before DC blocking. C3 then AC couples the signal to the external OUTA connection. R2 provides a 15 kΩ output load to ground, and TP7 provides access to the resulting AC-coupled output.

For Channel B, TP11 provides access to the direct output of U1B. C9 AC couples this signal to the external OUTB connection. R10 provides a 15 kΩ output load to ground, and TP12 provides access to the AC-coupled Channel B output.

This arrangement allows the user to observe both the internally biased amplifier signals and the corresponding zero-referenced AC output signals.

Amplifier Comparison

The two channels are intended to make the behavior of non-inverting and inverting operational-amplifier configurations easy to compare using a common input signal and common power supply.

Channel A produces an amplified signal with the same polarity as the input, while Channel B produces an amplified signal of approximately the same magnitude with opposite polarity. The available test points allow the user to observe how the signal changes as it passes through the input coupling, DC bias, feedback, amplification, and output coupling portions of each circuit.

Parts

Simulation in LTspice

LTspice was used to verify that selected component values produced reasonable amplifier behavior before assembling physical boards. The simulation helped confirm that the circuit was biased correctly, the signal was amplified, and the overall configuration functioned as expected with voltage gains approximately equal to those shown on the schematic. While the analysis wasn't exhaustive, the simulation served as a practical check to ensure the design was sound and worth building.

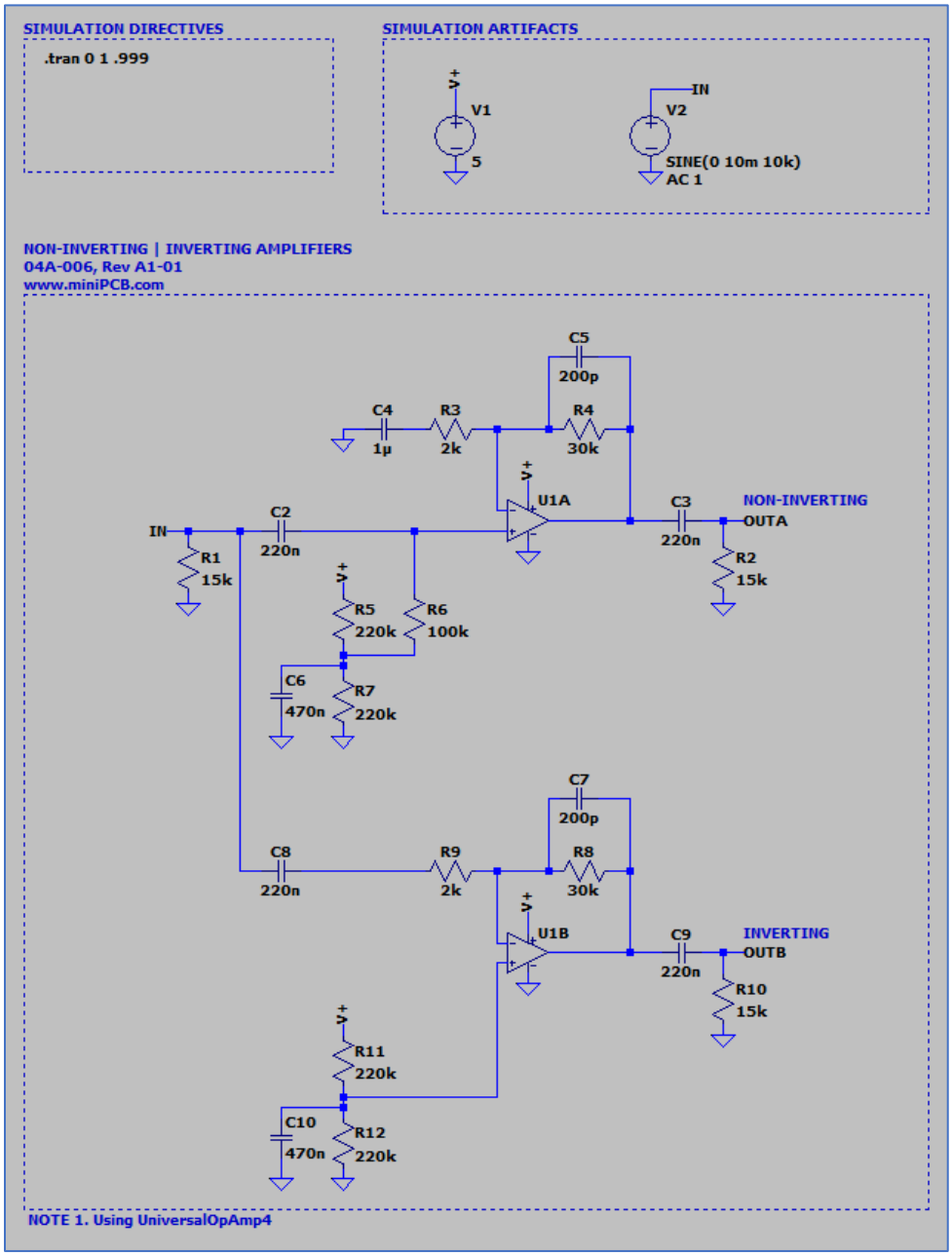


Figure 2 - LTspice Simulation, Schematic

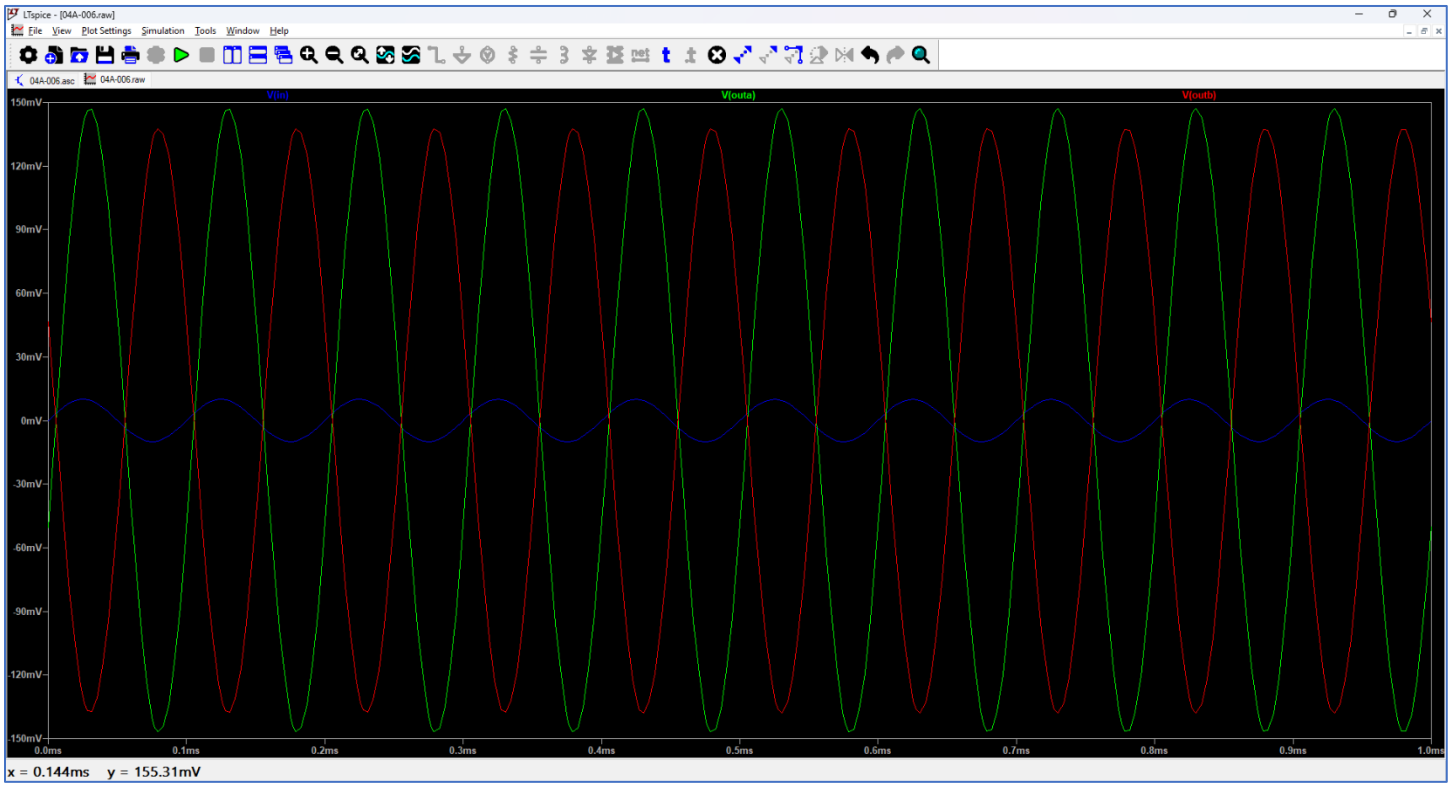


Figure 3 - LTspice Simulation, Signals

Example Gain Setting Resistor Values

Keep R3 and R9 fixed at 2k, then set these values for R4 and R8 to achieve different gains.

R2 and R8 (Ω)	Channel A Gain (V/V)	Channel B Gain (V/V)
2k	+2	-1
4.7k	+3.35	-2.35
10k	+6	-5
20k	+11	-10
30k, supplied values with kit	+16	-15
47k	+24.5	-23.5
100k	+51	-50

These are nominal midband gains. Actual gain depends on frequency and component tolerances. Larger gains reduce the input amplitude available before output clipping. Because C5 and C7 remain across the feedback resistors, changing resistance also changes the high-frequency response.

Example Parts List

REF DES	PART TYPE	VALUE / DESCRIPTION
C1	CAPACITOR	4.7uF
C2	CAPACITOR	220nF
C3	CAPACITOR	220nF
C4	CAPACITOR	1uF
C5	CAPACITOR	200pF
C6	CAPACITOR	470nF
C7	CAPACITOR	200pF
C8	CAPACITOR	220nF
C9	CAPACITOR	220nF
C10	CAPACITOR	470nF
R1	RESISTOR	15kΩ
R2	RESISTOR	15kΩ
R3	RESISTOR	2kΩ
R4	RESISTOR	30kΩ
R5	RESISTOR	220kΩ
R6	RESISTOR	100kΩ
R7	RESISTOR	220kΩ
R8	RESISTOR	30kΩ
R9	RESISTOR	2kΩ
R10	RESISTOR	15kΩ
R11	RESISTOR	220kΩ
R12	RESISTOR	220kΩ
U1	OPERATIONAL AMPLIFIER	TL072 dual operational amplifier
TP1-TP14	TEST POINT	KEYSTONE ELECTRONICS SERIES 5000
P1	HEADER PINS	6POS, 2.5mm PITCH, RA

Parts List Form

REF DES	PART TYPE	VALUE / DESCRIPTION
C1	CAPACITOR	
C2	CAPACITOR	
C3	CAPACITOR	
C4	CAPACITOR	
C5	CAPACITOR	
C6	CAPACITOR	
C7	CAPACITOR	
C8	CAPACITOR	
C9	CAPACITOR	
C10	CAPACITOR	
R1	RESISTOR	
R2	RESISTOR	
R3	RESISTOR	
R4	RESISTOR	
R5	RESISTOR	
R6	RESISTOR	
R7	RESISTOR	
R8	RESISTOR	
R9	RESISTOR	
R10	RESISTOR	
R11	RESISTOR	
R12	RESISTOR	
U1	OPERATIONAL AMPLIFIER	
TP1-TP14	TEST POINT	KEYSTONE ELECTRONICS SERIES 5000
P1	HEADER PINS	6POS, 2.5mm PITCH, RA

Build

To be continued.

Testing

To be continued.

Industry Feedback

Send your feedback to nolan@minipcb.com.

References

RESOURCE	LINK
This document	https://github.com/miniPCB/minipcb.github.io/blob/main/datasheets/04A-006.pdf
Web page	https://www.minipcb.com/04A/04A-006.html
Markdown	https://github.com/miniPCB/minipcb.github.io/blob/main/md/04A-006_A1-01_sch.md
JSON	https://github.com/miniPCB/minipcb.github.io/blob/main/json/04A-006_A1-01_sch.json
LTspice	https://github.com/miniPCB/minipcb.github.io/blob/main/ltspice/04A-006.asc
YouTube	https://www.youtube.com/@minipcb

Revision History

REV	DESCRIPTION	DATE
-	Initial release	20SEP2026
1	Added introduction about how components work Added references	20SEP2026
2	Added images from first article inspection Added LLM interaction	14OCT2026
3	Added example gain setting resistor values Corrected typos	16OCT2026